

Project 3 - DSA 8020

Saransh Rakshak

Due: Thursday May 1, 2025

Contents

Project Description	1
Problem 1. Analyzing Yearly Average Temperatures Time-Series	1
1. The following steps are needed to extract the annual mean temperatures at the specified location from the original dataset with daily temperatures at thousands of locations: .	1
2. Describe the main features of the annual time series data you extracted.	3
3. Is it reasonable to assume that there is a linear trend? If so, estimate and remove the trend to get the detrended time series.	4

Project Description

The ERA-Interim is a global atmospheric reanalysis dataset. Reanalysis refers to the process of producing spatially and temporally gridded datasets through data assimilation, primarily for climate monitoring and analysis. In the R data file **NA_MaxT_ERA.RData**, you will find daily maximum temperature values from 1979 to 2017, with a spatial resolution of approximately 80 km. The original global dataset has been subsetted to a region covering the contiguous United States and extending into Canada and Mexico.

In Problems 1 and 2, you are required to analyze an ERA-Interim average temperature time series at yearly and monthly temporal resolutions at a spatial location of your choice.

Problem 1. Analyzing Yearly Average Temperatures Time-Series

1. The following steps are needed to extract the annual mean temperatures at the specified location from the original dataset with daily temperatures at thousands of locations:

```
load("NA_MaxT_ERA.RData")
NA_MaxT <- NA_MaxT - 273.15 # change from Kelvin to Celsius
nlon <- length(NA_lon); nlat <- length(NA_lat)
nmon <- 12; nyr <- 39 # numbers of month/year
```

a. First, use the following R script to load the R data object:

```
# Clemson Example
Clemson.lon.lat <- c(-82.8374, 34.6834) # (lon,lat)
dist2Clemson <- rdist.earth(
  matrix(Clemson.lon.lat, 1, 2),
  expand.grid(NA_lon, NA_lat),
  miles = F
)
id <- which.min(dist2Clemson)
(lon_id <- id %% nlon)
```

b. Second, choose a location of interest (with the information of longitude and latitude) and identify the nearest ERA grid cell to the chosen location. Below you can find a script to get the correct ERA grid cell for Clemson:

```
## [1] 58
```

```
(lat_id <- id %% nlon + 1)
```

```
## [1] 15
```

```
# Check
(rdist.earth(matrix(Clemson.lon.lat, 1, 2),
matrix(c(NA_lon[lon_id], NA_lat[lat_id]), 1, 2), miles = F)
== min(dist2Clemson))
```

```
##      [,1]
## [1,] TRUE
```

```
Monterey.lon.lat <- c(-121.8979, 36.5973) # (lon,lat)
dist2Monterey <- rdist.earth(
  matrix(Monterey.lon.lat, 1, 2),
  expand.grid(NA_lon, NA_lat),
  miles = F
)

id <- which.min(dist2Monterey)

(lon_id <- id %% nlon)
```

```
## [1] 5
```

```
(lat_id <- id %% nlon + 1)
```

```
## [1] 18
```

```
(rdist.earth(
  matrix(Monterey.lon.lat, 1, 2),
  matrix(c(NA_lon[lon_id], NA_lat[lat_id]), 1, 2),
  miles = F
)== min(dist2Monterey))
```

```
##      [,1]
## [1,] TRUE
```

```
time_series_daily <- NA_MaxT[lon_id, lat_id,]
print("Daily Time Series summary: ")
```

c. Third, aggregate the daily data to the annual average time series. You could use the following script to achieve this:

```
## [1] "Daily Time Series summary: "
```

```
print(summary(time_series_daily))
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    4.173  13.178  14.816  14.864  16.539  24.845
```

```
time_series_yearly <- tapply(time_series_daily, list(yr), mean)
print("Yearly Time Series summary: ")
```

```
## [1] "Yearly Time Series summary: "
```

```
print(summary(time_series_yearly))
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    14.06   14.44   14.74   14.86   15.26   16.18
```

2. Describe the main features of the annual time series data you extracted.

```
# using package tsfeatures for feature autoextraction
yearly_features <- tsfeatures(time_series_yearly)
print(yearly_features)
```

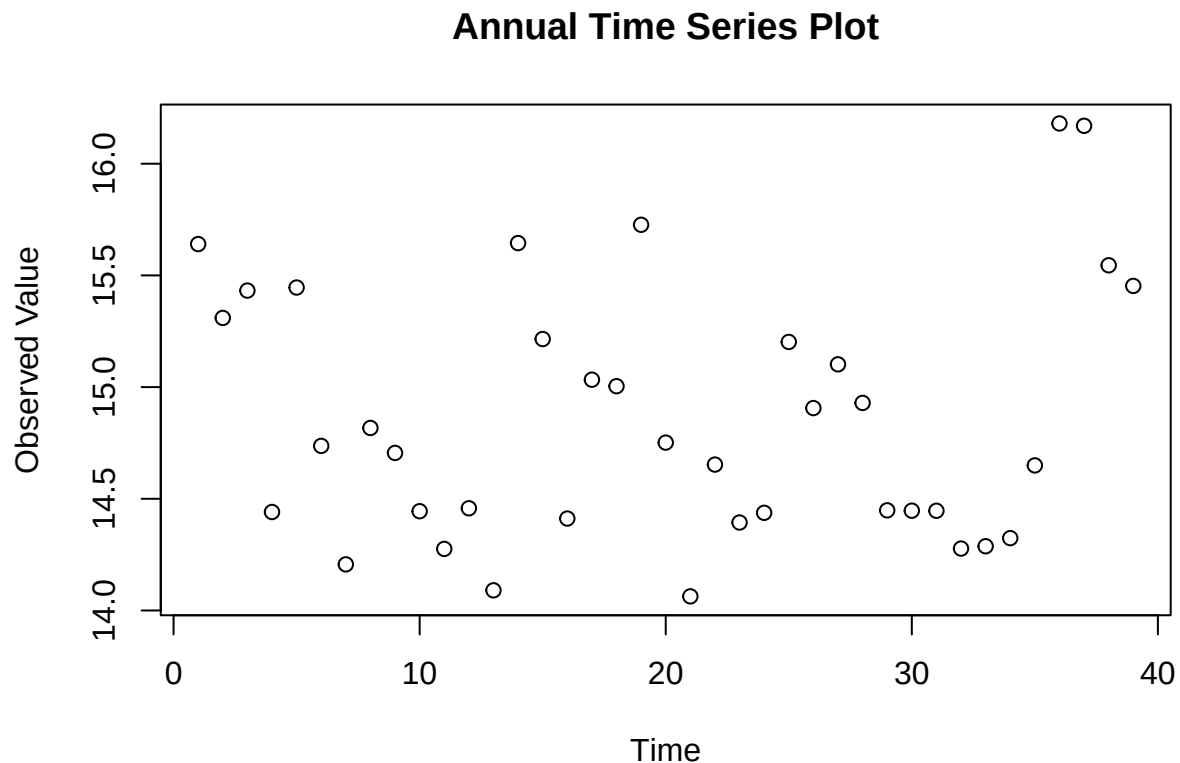
```
## # A tibble: 1 x 16
##   frequency nperiods seasonal_period trend      spike linearity curvature e_acf1
##   <dbl>     <dbl>         <dbl> <dbl>    <dbl>      <dbl>    <dbl> <dbl>
## 1         1         0             1 0.434 0.000433  0.0857    2.28 0.0822
## # i 8 more variables: e_acf10 <dbl>, entropy <dbl>, x_acf1 <dbl>,
## #   x_acf10 <dbl>, diff1_acf1 <dbl>, diff1_acf10 <dbl>, diff2_acf1 <dbl>,
## #   diff2_acf10 <dbl>
```

Our time series feature analysis using *tsfeatures* for the annual series reveals the following:

Feature	Value	Analysis
trend	0.4337	Moderately strong trend, indicates presence of long term pattern
spike	0.00042	Extremely low, indicates no significant anomalies or jumps in any given year
linearity	0.08572	Moderately low - some possibility model is linear
curvature	2.27771	High value strongly indicates non-linearity
entropy	0.9574	High value, indicates limited predictability and complex relations between the data
x_acf1	0.4117	Moderate indication of autocorrelation, any given years values show to depend on its previous year

3. Is it reasonable to assume that there is a linear trend? If so, estimate and remove the trend to get the detrended time series.

```
plot(time_series_yearly, main = "Annual Time Series Plot", ylab = "Observed Value", xlab = "Time")
```



Our graph does not reveal much significant information. However, based on our previous results from *tsfeatures*, it is still somewhat reasonable to expect the possibility of a linear trend:

1. trend: value 0.43 shows significant enough trend in data to expect some prediction.

2. linearity: while this is not 0, it is still low enough to keep linearity as a reasonable possibility
3. curvature: even though value is high (indicating a non-perfect linear fit), it is still significant

Since a linear trend is not out of question for our time series, we will now estimate its value.

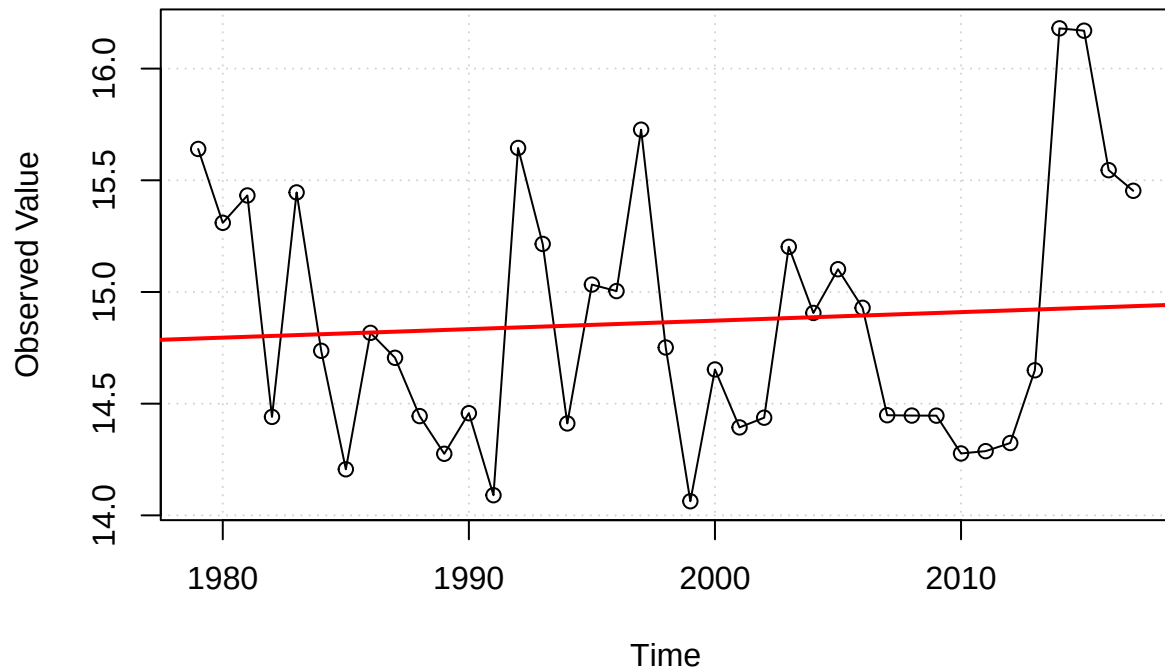
```
yrs <- unique(yr)

# fitting linear model
lm_annual <- lm(time_series_yearly ~ yrs)
summary(lm_annual)

##
## Call:
## lm(formula = time_series_yearly ~ yrs)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.8047 -0.4528 -0.1126  0.4420  1.2551
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  7.254522  16.272999   0.446   0.658
## yrs          0.003809   0.008145   0.468   0.643
##
## Residual standard error: 0.5724 on 37 degrees of freedom
## Multiple R-squared:  0.005876,    Adjusted R-squared:  -0.02099
## F-statistic: 0.2187 on 1 and 37 DF,  p-value: 0.6428

plot(
  x= yrs,
  y= time_series_yearly,
  type= "o", # show actual measured points with their continuity over time
  main = "Yearly Time Series(Trend)",
  ylab = "Observed Value",
  xlab = "Time",
  panel.first= grid()
)
abline(lm_annual, col = "red", lwd = 2)
```

Yearly Time Series(Trend)



Our linear model fit shows a p-value of 0.64, showing that a linear trend is statistically insignificant - leading us to fail to reject the null hypothesis and conclude there is no strong proof of linear behavior in the data. This is further confirmed by our Yearly Time Series(Trend) plot, which shows the data fluctuating significantly around our linear trend line (red) as well as many clearly non-linear patterns. It may be possible that the data could have a different measurable trend such as cyclical behavior, as is not unusual with environmental data.